Título: Cinemática directa — Robot 6GDL V9

Tabla DH (Craig / modificada) y robot

$x_{i-1}: z_{i-1} \to z_i$ $\quad$ $z_i: x_{i-1} \to x_i$

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	L_1	θ_1
2	$\pi/2$	0	0	$\theta_2 + \pi/2$
3	0	L_2	0	$\theta_3 - \pi/2$
4	$-\pi/2$	0	$L_3 + L_4$	θ_4
5	$\pi/2$	0	0	θ_5
6	$-\pi/2$	0	0	θ_6
7	0	0	$L_5 + L_{ef}$	0

$q = [\theta_1,\ \theta_2,\ \theta_3,\ \theta_4,\ \theta_5,\ \theta_6]^T$

Notación

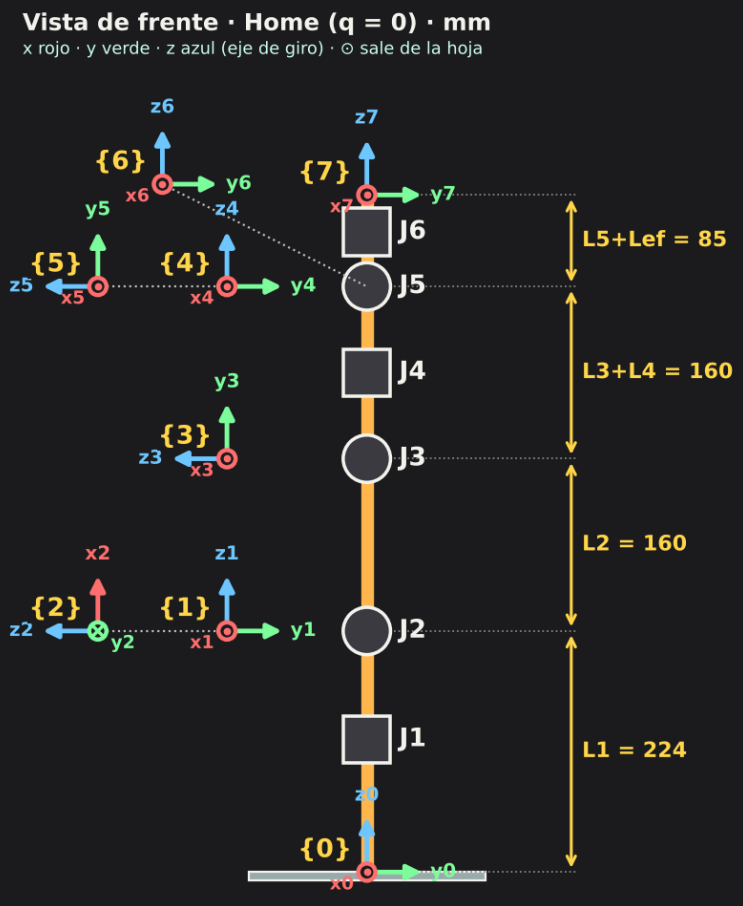
$c_i = \cos\theta_i, \quad s_i = \sin\theta_i$
$c_{23} = \cos(\theta_2 + \theta_3)$
$s_{23} = \sin(\theta_2 + \theta_3)$
$L_w = L_3 + L_4$ (hasta la muñeca)

Medidas (mm)

$L_1 = 224, \quad L_2 = 160$
$L_w = 160, \quad L_5 + L_{ef} = 85$

Muñeca esférica

Los ejes de J4, J5 y J6 se cruzan en el punto W, por eso ${}^4_5T$ y ${}^5_6T$ no tienen traslación.



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Matriz de una fila

$$ {}^{i-1}_{i}T = R_x(\alpha_{i-1})\, D_x(a_{i-1})\, R_z(\theta_i)\, D_z(d_i) = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ s\theta_i\, c\alpha_{i-1} & c\theta_i\, c\alpha_{i-1} & -s\alpha_{i-1} & -s\alpha_{i-1}\, d_i \\ s\theta_i\, s\alpha_{i-1} & c\theta_i\, s\alpha_{i-1} & c\alpha_{i-1} & c\alpha_{i-1}\, d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

Ojo con los offsets (identidades)

Fila 2: $\cos(\theta_2 + \frac{\pi}{2}) = -s_2, \quad \sin(\theta_2 + \frac{\pi}{2}) = c_2$

Fila 3: $\cos(\theta_3 - \frac{\pi}{2}) = s_3, \quad \sin(\theta_3 - \frac{\pi}{2}) = -c_3$

Con $\alpha = \pm\pi/2$: $\cos\alpha = 0$ y $\sin\alpha = \pm 1$, por eso aparecen los 0 y ± 1 en las columnas 2 y 3.

1. Matrices de cada fila

$$ {}^{0}_{1}T = \begin{bmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & L_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad {}^{1}_{2}T = \begin{bmatrix} -s_2 & -c_2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ c_2 & -s_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^{2}_{3}T = \begin{bmatrix} s_3 & c_3 & 0 & L_2 \\ -c_3 & s_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad {}^{3}_{4}T = \begin{bmatrix} c_4 & -s_4 & 0 & 0 \\ 0 & 0 & 1 & L_w \\ -s_4 & -c_4 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^{4}_{5}T = \begin{bmatrix} c_5 & -s_5 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ s_5 & c_5 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad {}^{5}_{6}T = \begin{bmatrix} c_6 & -s_6 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -s_6 & -c_6 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^{6}_{7}T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & (L_5 + L_{ef}) \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

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2. Multiplicación paso a paso

Se va multiplicando por la derecha: cada nueva fila se pega a lo que ya se tenía.

$$ {}^0_2T = {}^0_1T\ {}^1_2T = \begin{bmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & L_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -s_2 & -c_2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ c_2 & -s_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^0_2T = \begin{bmatrix} -c_1 s_2 & -c_1 c_2 & s_1 & 0 \\ -s_1 s_2 & -c_2 s_1 & -c_1 & 0 \\ c_2 & -s_2 & 0 & L_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^0_3T = {}^0_2T\ {}^2_3T = \begin{bmatrix} -c_1 s_2 & -c_1 c_2 & s_1 & 0 \\ -s_1 s_2 & -c_2 s_1 & -c_1 & 0 \\ c_2 & -s_2 & 0 & L_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} s_3 & c_3 & 0 & L_2 \\ -c_3 & s_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^0_3T = \begin{bmatrix} c_1 c_{23} & -c_1 s_{23} & s_1 & -L_2 c_1 s_2 \\ c_{23} s_1 & -s_1 s_{23} & -c_1 & -L_2 s_1 s_2 \\ s_{23} & c_{23} & 0 & L_1 + L_2 c_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

Aquí aparece c_{23}: $c_2 c_3 - s_2 s_3 = \cos(\theta_2 + \theta_3)$ y $s_2 c_3 + c_2 s_3 = \sin(\theta_2 + \theta_3)$ (J2 y J3 son paralelos).

$$ {}^0_4T = {}^0_3T\ {}^3_4T = \begin{bmatrix} c_1 c_{23} & -c_1 s_{23} & s_1 & -L_2 c_1 s_2 \\ c_{23} s_1 & -s_1 s_{23} & -c_1 & -L_2 s_1 s_2 \\ s_{23} & c_{23} & 0 & L_1 + L_2 c_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c_4 & -s_4 & 0 & 0 \\ 0 & 0 & 1 & L_w \\ -s_4 & -c_4 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^0_4T = \begin{bmatrix} c_1 c_{23} c_4 - s_1 s_4 & -c_1 c_{23} s_4 - c_4 s_1 & -c_1 s_{23} & -L_2 c_1 s_2 - L_w c_1 s_{23} \\ c_1 s_4 + c_{23} c_4 s_1 & c_1 c_4 - c_{23} s_1 s_4 & -s_1 s_{23} & -L_2 s_1 s_2 - L_w s_1 s_{23} \\ c_4 s_{23} & -s_{23} s_4 & c_{23} & L_1 + L_2 c_2 + L_w c_{23} \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

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${}^{0}_{5}T$ (se agrega J5)

$$
{}^{0}_{5}T = {}^{0}_{4}T\ {}^{4}_{5}T = \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$
\begin{aligned}
r_{11} &= c_1 c_{23} c_4 c_5 - c_1 s_{23} s_5 - c_5 s_1 s_4 \\
r_{12} &= -c_1 c_{23} c_4 s_5 - c_1 c_5 s_{23} + s_1 s_4 s_5 \\
r_{13} &= c_1 c_{23} s_4 + c_4 s_1 \\
p_x &= -L_2 c_1 s_2 - L_w c_1 s_{23} \\
r_{21} &= c_1 c_5 s_4 + c_{23} c_4 c_5 s_1 - s_1 s_{23} s_5 \\
r_{22} &= -c_1 s_4 s_5 - c_{23} c_4 s_1 s_5 - c_5 s_1 s_{23} \\
r_{23} &= -c_1 c_4 + c_{23} s_1 s_4 \\
p_y &= -L_2 s_1 s_2 - L_w s_1 s_{23} \\
r_{31} &= c_{23} s_5 + c_4 c_5 s_{23} \\
r_{32} &= c_{23} c_5 - c_4 s_{23} s_5 \\
r_{33} &= s_{23} s_4 \\
p_z &= L_1 + L_2 c_2 + L_w c_{23}
\end{aligned}
$$

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${}^{0}_{6}T$ (se agrega J6)

$$ {}^{0}_{6}T = {}^{0}_{5}T\ {}^{5}_{6}T = \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ r_{11} = c_1 c_{23} c_4 c_5 c_6 - c_1 c_{23} s_4 s_6 - c_1 c_6 s_{23} s_5 - c_4 s_1 s_6 - c_5 c_6 s_1 s_4 $$
$$ r_{12} = -c_1 c_{23} c_4 c_5 s_6 - c_1 c_{23} c_6 s_4 + c_1 s_{23} s_5 s_6 - c_4 c_6 s_1 + c_5 s_1 s_4 s_6 $$
$$ r_{13} = -c_1 c_{23} c_4 s_5 - c_1 c_5 s_{23} + s_1 s_4 s_5 $$
$$ p_x = -L_2 c_1 s_2 - L_w c_1 s_{23} $$
$$ r_{21} = c_1 c_4 s_6 + c_1 c_5 c_6 s_4 + c_{23} c_4 c_5 c_6 s_1 - c_{23} s_1 s_4 s_6 - c_6 s_1 s_{23} s_5 $$
$$ r_{22} = c_1 c_4 c_6 - c_1 c_5 s_4 s_6 - c_{23} c_4 c_5 s_1 s_6 - c_{23} c_6 s_1 s_4 + s_1 s_{23} s_5 s_6 $$
$$ r_{23} = -c_1 s_4 s_5 - c_{23} c_4 s_1 s_5 - c_5 s_1 s_{23} $$
$$ p_y = -L_2 s_1 s_2 - L_w s_1 s_{23} $$
$$ r_{31} = c_{23} c_6 s_5 + c_4 c_5 c_6 s_{23} - s_{23} s_4 s_6 $$
$$ r_{32} = -c_{23} s_5 s_6 - c_4 c_5 s_{23} s_6 - c_6 s_{23} s_4 $$
$$ r_{33} = c_{23} c_5 - c_4 s_{23} s_5 $$
$$ p_z = L_1 + L_2 c_2 + L_w c_{23} $$

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${}^{0}_{7}T$ — base {0} → punta {7}

$$ {}^{0}_{7}T = {}^{0}_{6}T\ {}^{6}_{7}T = \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$
\begin{aligned}
r_{11} &= c_1c_{23}c_4c_5c_6 - c_1c_{23}s_4s_6 - c_1c_6s_{23}s_5 - c_4s_1s_6 - c_5c_6s_1s_4 \\
r_{12} &= -c_1c_{23}c_4c_5s_6 - c_1c_{23}c_6s_4 + c_1s_{23}s_5s_6 - c_4c_6s_1 + c_5s_1s_4s_6 \\
r_{13} &= -c_1c_{23}c_4s_5 - c_1c_5s_{23} + s_1s_4s_5 \\
p_x &= -L_2c_1s_2 + (L_5 + L_{ef})\,(-c_1c_{23}c_4s_5 - c_1c_5s_{23} + s_1s_4s_5) - L_wc_1s_{23} \\
r_{21} &= c_1c_4s_6 + c_1c_5c_6s_4 + c_{23}c_4c_5c_6s_1 - c_{23}s_1s_4s_6 - c_6s_1s_{23}s_5 \\
r_{22} &= c_1c_4c_6 - c_1c_5s_4s_6 - c_{23}c_4c_5s_1s_6 - c_{23}c_6s_1s_4 + s_1s_{23}s_5s_6 \\
r_{23} &= -c_1s_4s_5 - c_{23}c_4s_1s_5 - c_5s_1s_{23} \\
p_y &= -L_2s_1s_2 + (L_5 + L_{ef})\,(-c_1s_4s_5 - c_{23}c_4s_1s_5 - c_5s_1s_{23}) - L_ws_1s_{23} \\
r_{31} &= c_{23}c_6s_5 + c_4c_5c_6s_{23} - s_{23}s_4s_6 \\
r_{32} &= -c_{23}s_5s_6 - c_4c_5s_{23}s_6 - c_6s_{23}s_4 \\
r_{33} &= c_{23}c_5 - c_4s_{23}s_5 \\
p_z &= L_1 + L_2c_2 + (L_5 + L_{ef})\,(c_{23}c_5 - c_4s_{23}s_5) + L_wc_{23}
\end{aligned}
$$

Comprobación en Home ($q = 0$)

Con $c_i = 1,\ s_i = 0,\ c_{23} = 1,\ s_{23} = 0$: $p_x = 0,\quad p_y = 0,\quad p_z = L_1 + L_2 + L_w + (L_5 + L_{ef}) = 224 + 160 + 160 + 85 =$ **629** mm ✓

Coincide con SolidWorks, Simulink, MATLAB y el programa del profe.